Emerging Liability Severity Under Sparse Claims: Disclosure Censoring and the Identifiability Frontier with an Illustrative Application to AI Systems

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Abstract

Pricing emerging liability risks before a mature claims history exists is a recurring actuarial problem, from catastrophe modelling to cyber. Public incident data for such risks are sparse and *disclosure-censored*: only losses above an effective reporting threshold enter the public record, so a severity estimate must borrow strength from mature proxy lines. We frame the question as one of *identifiability*: borrowing from a proxy through a commensurate prior, *when does a data-poor line's severity posterior transition from prior-driven to data-driven?* We make three contributions. (i) We argue that loss frequency is structurally non-identifiable from public incident catalogues (no exposure denominator), so an emerging-risk analysis is necessarily severity-first. (ii) We define an *identifiability frontier* $n^\star$ at which the posterior escapes the prior, recover the classical Bühlmann–Straub factor as its conjugate special case, and show that left-truncation deflates the effective sample size by a factor $1-\delta(\alpha)$, inflating $n^\star$ by $1/(1-\delta(\alpha))$ (Lemma 1, Corollary 1). (iii) We apply the framework to a public-record screen of AI-attributed losses. The empirical result is negative: a screening funnel and an illustrative case set show that the prior-driven versus data-driven verdict is *not identified* from currently available public data, because it flips across plausible prior strengths. The methodological components are recombinations of known results (credibility theory, the truncated-normal information, the lognormal limited expected value); the contribution is the framing and the empirical localisation, not a new theorem.

Keywords: credibility theory; commensurate priors; left-truncation; effective sample size; emerging risk; AI liability.

1 Introduction

Actuaries repeatedly price novel risks before a mature claims ledger exists, commercial aviation, catastrophe aggregation, cyber, and now artificial-intelligence (AI) systems. A standalone market for affirmative AI-liability cover has emerged (Lloyd's coverholders, Munich Re's aiSure, and specialist managing general agents), priced today on bespoke, proprietary bases. There is, however, no published, data-calibrated method for the *severity* of AI-attributed third-party economic losses.

Two structural features make naive severity estimation infeasible for emerging risks. First, *exposure is unmeasured*: public incident catalogues are global numerators with no denominator, so frequency (claims per exposure unit) is not identifiable from public data, and loss cost in the textbook sense (frequency × severity) cannot be formed. We therefore restrict attention to severity and state frequency non-identifiability as a finding. Second, severity data are *sparse and disclosure-censored*: only losses crossing a reporting threshold (regulatory materiality, court-filing minima, news salience) enter the record.

In the absence of own-line data the instrument is *credibility*: borrow from a related, mature proxy and let data correct the prior. The question we make precise is not "what is the expected loss" but "how much of any answer is data and how much is the borrowed prior."

Contributions and scope.

1. **Frequency non-identifiability** from public catalogues, motivating a severity-first analysis.

2. **An identifiability frontier** $n^\star(t)$ deflated by disclosure-censoring (Lemma 1, Corollary 1), with a worked layer-pricing implication (Proposition 1). The underlying pieces, Bühlmann–Straub credibility, the truncated-normal Fisher information, the lognormal limited expected value, are standard; the synthesis is the contribution.

3. **An illustrative empirical localisation** on a public-record AI screen, reported with its limitations, showing the regime verdict is *not identified* from current data.

We do not claim a new theorem, a verified loss dataset, or a deployable rate.

2 Related work

Emerging-risk precedents. Pricing sparse, unobserved tails recurs in the literature: catastrophe modelling shifted from experience rating to engineering priors [Clark, 2002], and cyber exposed the limits of credibility borrowing under accumulation [Biener et al., 2015]. We add an explicit frontier for *when* proxy-pricing should yield to data-pricing.

AI insurance (qualitative/architectural). Four 2026 contributions treat AI insurance without calibrating a severity distribution from loss data: Zhu [2026] (a conceptual framework), Leung et al. [2026a] (a coverage taxonomy, the *insurability* frontier), Chen [2026] (runtime control, an *authority* frontier), and Leung et al. [2026b] (a loss-reconstruction diagnostic). Our *identifiability* frontier is distinct from these.

Credibility, borrowing, truncation. Exact credibility is due to Jewell [1974]; Diaconis & Ylvisaker [1979] characterise conjugacy by posterior-mean linearity; Bühlmann & Gisler [2005] is standard. Dynamic borrowing, commensurate [Hobbs et al., 2011], power [Ibrahim & Chen, 2000, Han et al., 2022], robust-MAP [Schmidli et al., 2014], repeatedly finds the borrowing strength weakly identified at small n. Left-truncated and trimmed loss models are standard [Klugman et al., 2019, Kim & Jeon, 2013].

3 Model and inclusion boundary

Severity scale. Let $Y = \log X$, $Y \sim \mathcal{N}(\theta, \sigma^2)$ with σ the within-line dispersion, and a commensurate proxy prior $\theta \sim \mathcal{N}(m_0, \tau^2)$. We model the *location* θ; the dispersion σ (the tail) is held at the proxy value, a restriction we return to as a central limitation.

Inclusion boundary. We target third-party *economic-loss* liability from an AI system in a professional/commercial-service role (advice, information, content, customer interaction, decision support), excluding bodily injury/property and systemic financial-market events. Operationalising this boundary on public data is hard, and our illustrative sample does not yet honour it cleanly (Section 7).

Loss-type strata. Indemnities, settlements, awards, fines and demands are distinct random variables and must not be pooled. The intended estimand is the paid indemnity/settlement.

4 Credibility and the identifiability frontier

Define

$$Z \;=\; 1 - \frac{\mathrm{Var}(\theta \mid \mathrm{data})}{\mathrm{Var}(\theta \mid \mathrm{prior})}, \tag{1}$$

the fraction of prior variance resolved by data; $Z = \frac{1}{2}$ is the prior-/data-driven boundary. In the untruncated conjugate case, posterior precision $1/\tau^2 + n/\sigma^2$ gives

$$Z \;=\; \frac{n}{n+k}, \qquad k = \frac{\sigma^2}{\tau^2}, \tag{2}$$

which coincides with the Bühlmann–Straub credibility factor [Jewell, 1974, Bühlmann & Gisler, 2005].

Remark 1 (Z is a variance-reduction ratio)**.** *Equation* (2) *equals the Bühlmann–Straub data weight only in the conjugate case. Once conjugacy fails (Section 5), the variance-reduction ratio and the posterior-mean weight diverge, and the truncated sample mean is biased (*$E[\bar{Y} \mid Y > t] = \theta + \sigma\lambda(\alpha)$*). We therefore treat Z below as an information statistic, not a posterior-mean weight, and flag the truncation bias as a limitation rather than correcting it here.*

5 Effective sample size under disclosure-censoring

Severity data are left-truncated at a disclosure threshold t: Y observed iff $Y > t$. The truncated-normal likelihood is non-conjugate, so exact linear credibility fails.

Lemma 1 (Truncated-normal information deflation; standard)**.** *For $Y \sim \mathcal{N}(\theta, \sigma^2)$ observed on $Y > t$, with $\alpha = (t-\theta)/\sigma$ and $\lambda(\alpha) = \phi(\alpha)/(1-\Phi(\alpha))$, the per-observation Fisher information for θ is $\mathcal{I}(\theta; t) = (1-\delta(\alpha))/\sigma^2$, $\delta(\alpha) = \lambda(\alpha)(\lambda(\alpha) - \alpha) \in [0, 1)$.*

Proof. The score is $\partial_\theta \log f = (y-\theta)/\sigma^2 - \lambda(\alpha)/\sigma$; the second term is constant in y, and $\mathrm{Var}(Y \mid Y > t) = \sigma^2(1-\delta(\alpha))$ is the standard truncated-normal variance [Klugman et al., 2019], so $\mathcal{I} = \mathrm{Var}(Y \mid Y > t)/\sigma^4 = (1-\delta(\alpha))/\sigma^2$. □

A Laplace approximation with precision $1/\tau^2 + n\,\mathcal{I}(\theta; t)$ gives, evaluated at $\hat{\theta}$,

$$n_{\mathrm{eff}} = n\,(1-\delta(\alpha)), \qquad Z_{\mathrm{eff}} = \frac{n_{\mathrm{eff}}\,\tau^2}{n_{\mathrm{eff}}\,\tau^2 + \sigma^2}, \tag{3}$$

recovering (2) as $t \to -\infty$. Because α depends on θ, $\delta(\alpha)$ is a plug-in evaluated at $\hat{\theta}$, not a design constant.

Corollary 1 (Frontier inflation)**.** *$Z_{\mathit{eff}} = \frac{1}{2} \iff n^\star(t) = (\sigma^2/\tau^2)/(1-\delta(\alpha))$: censoring inflates the data requirement by $1/(1-\delta(\alpha))$.*

Numerical validation. Against a deterministic grid posterior under the truncated likelihood ($\theta = 13.5, \sigma = 2, m_0 = 13.5, \tau = 0.8$, $k = 6.25$), the closed form (3) agrees to a maximum absolute error of 0.006 across $n \in \{5, 10, 20, 40\}$ and truncation quantiles $\{0, 0.25, 0.5, 0.75\}$, and reproduces (2) exactly with no truncation (reproducible code released). Figure 1 plots the resulting frontier.

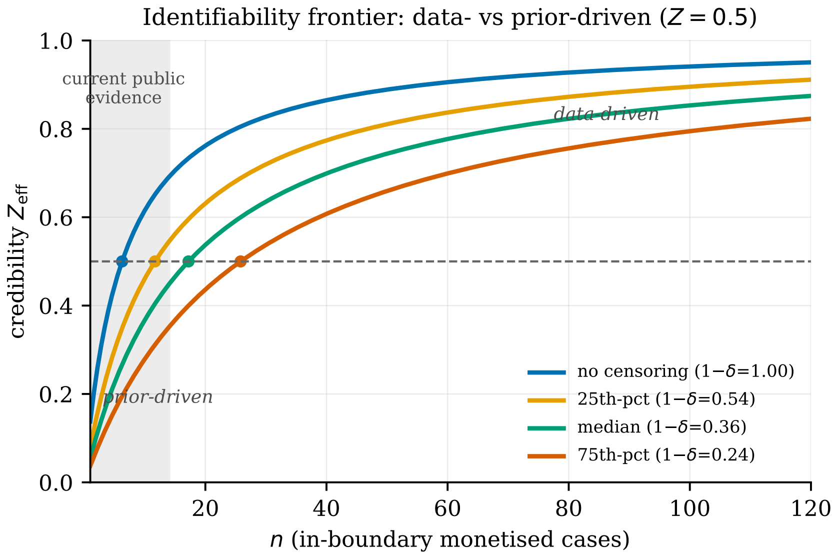

Figure 1: The identifiability frontier. Credibility Z_{eff} versus in-boundary sample size n for several disclosure-censoring levels ($k = \sigma^2/\tau^2 = 6.25$). The $Z = \frac{1}{2}$ line separates the prior-driven and data-driven regimes; markers show $n^\star = k/(1-\delta)$. Heavier censoring pushes $n^\star$ outward; currently available public in-boundary evidence (shaded) sits in the prior-driven region.

Proposition 1 (Layer-pricing implication; location only)**.** *For a lognormal severity the limited expected value of a layer attaching at A with limit L follows the standard closed form [Klugman et al., 2019]; substituting the credibility location $\mu_{post} = m_0 + Z_{eff}(\hat{\theta} - m_0)$ and predictive variance $\sigma^2_{pred} = \sigma^2 + \tau^2(1 - Z_{eff})$ shows the layer premium is updated by data only through the location. Because σ (hence the tail) is held at the proxy, upper excess layers ($A \gg e^{m_0}$) are insensitive to data by construction, not as a consequence of the frontier; only working layers near $e^{\mu_{post}}$ respond to data.*

Proposition 1 is therefore a caution, not a pricing tool for excess layers: a credible excess-layer treatment requires updating the tail, which we do not do here.

Sensitivity of the frontier. Table 1 tabulates $n^\star = k/(1-\delta)$. The frontier moves by 2–4× with censoring and by the same order with prior strength k; it is not a single number.

6 Proxy-line priors

We illustrate prior calibration from published cyber/operational-risk severity statistics (Table 2), matching a lognormal by $\mu = \log(\text{median})$, $\sigma = \sqrt{2(\log \text{mean} - \log \text{median})}$. *This moment-match is unreliable when the proxy is heavy/infinite-mean tailed* [Eling & Jung, 2022, Cremer et al., 2023]; we use it only to locate a plausible k, not as a tail model, and note the unresolved tension between a lognormal location model and the heavy tails the proxy exhibits.

Censoring	$1-\delta(\alpha)$	**Required $n^\star$ by prior strength $k=\sigma^2/\tau^2$** Strong (k=10)	Baseline (k=6.25)	Weak (k=4)	Diffuse (k=2)
none	1.000	10.0	6.2	4.0	2.0
25th pct	0.535	18.7	11.7	7.5	3.7
median	0.363	27.5	17.2	11.0	5.5
75th pct	0.242	41.3	25.8	16.5	8.3

Table 1: Frontier $n^\star$ across censoring and (weakly identified) prior strength k. Both axes are assumptions, not estimates; the regime verdict depends on the cell chosen.

proxy sub-line	median ($M)	mean ($M)	source
Privacy: unauthorized disclosure	2.92	9.49	Malavasi et al. [2021]
Data: malicious breach	0.94	28.96	Malavasi et al. [2021]
Identity: fraudulent use	0.16	1.83	Malavasi et al. [2021]
Cyber aggregate	0.11	16.84	Cremer et al. [2023]
Financial op-risk (> $100k)	1.10	24.95	Eling & Jung [2022]

Table 2: Published proxy severity statistics (different units of observation; pooling is itself an approximation). The $100k floor is a disclosure threshold, motivating Section 5.

7 Illustrative empirical localisation

Screen. We re-screen *all* 1,505 AIID incidents (CC-BY-SA 4.0) against the Section 3 inclusion boundary, not only those matching a monetary regex, which miss small or foreign-currency figures (e.g. the canonical *Moffatt v. Air Canada* chatbot case). Out-of-boundary classes (wrongful-arrest, biometric/BIPA, physical-safety, employment, government-benefit) are excluded with documented per-case reasons; the screen is released as auditable code and CSV.

Composition of the in-boundary set. The re-screen returns 181 in-boundary candidates. Their composition: 75 are copyright/IP matters carrying *unpaid* demands (multi-billion-dollar prayers for relief in training-data suits), 35 are regulatory fines, 16 are lawyer-sanction matters (AI-fabricated citations), and only ∼28 are genuine "AI service → third-party economic loss". Of those, the cases with a confirmed *paid* figure number only a handful and span four orders of magnitude, from the Air Canada chatbot tribunal award (≈CAD$812) to AI-defamation settlements of $5M (Meta) and $15M (Google), and are dominated by defamation rather than the advisory/decision-support losses the boundary targets. Pooling the headline figures of distinct legal regimes (copyright demands, fines, sanctions, settlements), as a naive screen does, both mis-classifies and overstates the severity sample (Figures 2–3).

Defects of the in-boundary screen. Even the in-boundary screen has material defects, each of which alone undermines a pricing inference: (a) figures are secondary, mixed-currency, nominal across ∼2015–2025 and not yet confirmed against court/regulator primary sources; (b) loss types (settlement / award / fine / demand) are not commensurable and must not be pooled; (c) the sample is *right*-censored by policy limits and ability-to-pay (settlements cluster at tower limits, not true harm), compressing $\hat{\sigma}$, and this is unmodelled; (d) the disclosure mechanism is selection-on-unobservables (large losses settle under NDA *to avoid* disclosure), violating the fixed-threshold

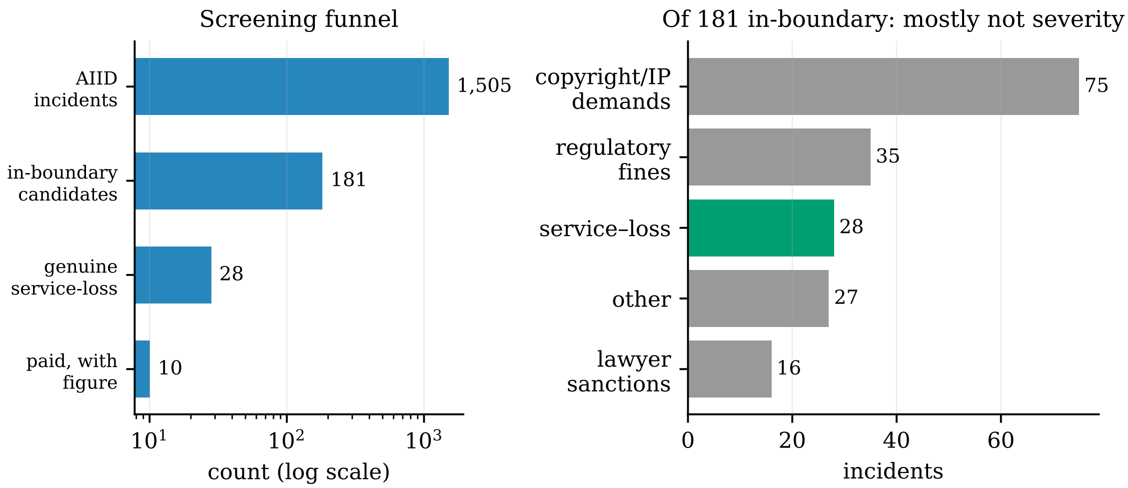

Figure 2: Screening funnel (left) and category composition of the 181 in-boundary candidates (right). The overwhelming majority are copyright/IP demands, regulatory fines, or lawyer-sanction matters; only ∼28 are genuine "AI service → third-party economic loss", and fewer still carry a paid figure.

left-truncation the model assumes; (e) recent cases are immature (IBNR); (f) AIID is itself curated for notable incidents, a second selection layer away from an underwriter's exposure population.

What can be said. With only a handful of genuinely in-boundary paid settlements, the severity *location* cannot be estimated with any precision, let alone localised on the frontier of Table 1: any Z_{eff} computed from such an n is dominated by sampling and classification noise far exceeding any plausible regime margin, and the verdict flips across plausible prior strengths k (which is itself weakly identified). **The prior-driven-versus-data-driven regime is therefore not identified from current public data.** That negative result, public AI-liability loss experience is too sparse, heterogeneous, and selection-distorted to displace a proxy prior, is the paper's empirical conclusion. A severity *calibration* would require non-public insurer claims data; public catalogues can support a screen and an identifiability statement, not a rate.

8 Limitations and threats to validity

(i) **Frequency** is non-identifiable from public data (a result, not an omission). (ii) **Location only**: the result updates θ, not the tail σ; the loss cost is tail-driven, so excess-layer pricing is out of scope (Proposition 1). (iii) **Independence**: the model assumes i.i.d. severities, but AI failures are correlated/fleet-wide (one model defect → many simultaneous claims), an accumulation risk the i.i.d. credibility machinery and the idiosyncratic cyber proxy both ignore. (iv) **Proxy isomorphism** is assumed, not defended: cyber severity scales with records breached, AI economic loss with downstream reliance on wrong outputs, different tail drivers. (v) **Heavy tails**: the lognormal location model contradicts the heavy/infinite-mean cyber tails cited; moment-matching σ is unreliable there. (vi) **Data**: right-censoring, selection-on-unobservables, IBNR immaturity, currency/year mixing, and boundary violations (above) make the empirical set illustrative only. (vii) **Scope of "agent"**: the AIID-era cases are pre-agentic (advisory/content tools), so claims

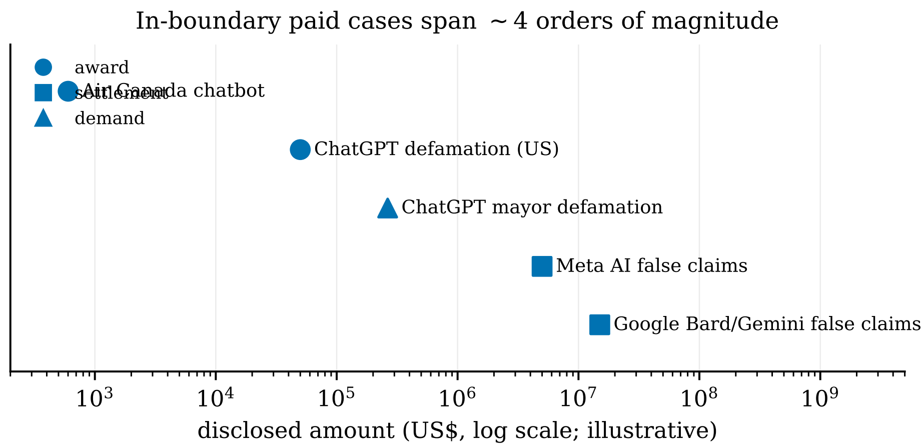

Figure 3: Illustrative in-boundary paid cases on a log-dollar scale (amounts approximate, pending primary-source verification). The span of roughly four orders of magnitude, and the dominance of defamation, illustrates why a severity distribution cannot be calibrated from this sample.

about autonomous AI *agents* are not supported by this sample. (viii) **Truncation bias**: $\bar{Y} \mid Y > t$ overstates θ; a Heckman/Tobit correction is deferred.

9 Conclusion

Framing emerging-risk severity as an identifiability question, and giving an explicit, censoring-inflated frontier $n^\star(t)$, is the applied contribution, and the empirical finding is a *negative* one: with current public data the prior-/data-driven regime for AI-liability severity is not identified. The mathematical components are standard recombinations, not new theorems; the contribution is the framing and the empirical localisation. Realistic venue is an applied actuarial outlet (e.g. *Variance* or *Risks*). Future work that would make this a substantive contribution: a sourced, in-boundary, censoring- and selection-corrected dataset; an accumulation/dependence model for fleet-correlated AI failures; and extension of the credibility update from the location to the tail and the loss-cost functional.

Reproducibility. Code, proxy priors, the truncation validation, and the screening protocol are at `https://github.com/tayyub-ai/actuarial-loss-cost-modelling-for-ai-agent-liability`. The illustrative case set is *not* a verified dataset and is labelled as such.

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